
CONTEXTUAL NORMAL RETURN FIELDS FOR POINT-LEVEL LiDAR ANOMALY SEGMENTATION

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ABSTRACT

Point-level LiDAR anomaly segmentation requires a model to recognize objects beyond its training vocabulary while retaining normal points under sparse and unfamiliar observations. Auxiliary anomalies provide discriminative supervision, and normal-scene prediction offers a complementary description of expected observations. We introduce contextual normal return fields, a representation that connects these two sources of supervision through the geometry of LiDAR measurements. A context encoder excludes a target angular block and predicts multiple three-dimensional return-rate fields shared by its rays. Analytic integration converts each field into a conditional distribution over return distance. A hierarchical compatibility decoder compares the observed point with individual field components and complete normal hypotheses, producing one anomaly score per point. The model learns jointly from point-level anomaly labels and a shared-hypothesis likelihood on normal returns. Training data are constructed entirely from nuScenes through observation-supported object insertion, with normal objects inserted by the same procedure as auxiliary anomalies. Evaluation is organized around AP, AUROC, and FPR95 on STU, together with controlled comparisons of the normal representation and its readout.

[Quantitative findings to be inserted after evaluation.]

1 INTRODUCTION

An unfamiliar object can occupy only a handful of LiDAR returns. Identifying these points requires a model to separate semantic novelty from the variation produced by range, sampling density, and scene layout. A distant pedestrian remains normal even when its shape is poorly resolved; an unfamiliar object can resemble part of a known vehicle. The point-level anomaly segmentation task captures this distinction by assigning an anomaly score to each observed return. STU provides a driving benchmark in which the target objects lie outside a specified set of normal semantic categories (Nekrasov et al., 2025).

Existing approaches obtain anomaly evidence from semantic representations, auxiliary unknown examples, and learned normal distributions. Open-set LiDAR models expose a segmentation network to transformed or inserted objects (Cen et al., 2022; Xu et al., 2025), learn anomaly-aware scores (Li et al., 2026c;b), or compare observations with normal semantic prototypes (Mosco et al., 2026). Generative approaches reconstruct normal features and use their discrepancy from the input as additional evidence (Oh et al., 2026). The quality of a normal reference therefore matters: it determines which aspects of an observation can support an anomaly decision beyond resemblance to the auxiliary objects encountered during training.

We study a normal reference that is constrained by both its input information and its observation geometry. When predicting a target region, the reference uses measurements outside that region. Within the region, several rays query a common three-dimensional hypothesis. This construction links their predicted distances through shared geometry. Multiple hypotheses represent alternative normal arrangements supported by the surrounding scene. The resulting distributions remain available to the anomaly decoder at inference, where they can be compared with the actual returns.

This leads to *contextual normal return fields*. For each angular block, a context encoder predicts a mixture of spatial return-rate fields. Integrating a field along an observed ray produces a normalized distribution over return distance. During normal learning, one mixture component jointly

explains the valid normal returns in the block. During anomaly inference, the observed point reads the ray-specific parameters of these hypotheses through a hierarchical compatibility decoder. An actual-observation encoder supplies scene and point detail, and a single output head combines these representations into the anomaly score.

The distinction between normal prediction and anomaly judgment is central to this design. Normal density describes support under a learned distribution, while anomaly labels specify semantic membership (Nalisnick et al., 2019; Le Lan & Dinh, 2021). Normal observations can have several plausible continuations, while auxiliary anomalies sample a finite set of unknown objects. A learned decoder uses the predicted distribution together with the observed point to resolve the task’s semantic decision. This follows the broader motivation for combining normal modeling with discriminative learning (Grcić et al., 2022; Zavrtanik et al., 2021), and develops it around a shared spatial representation of LiDAR returns.

Our training construction gives each learning objective a corresponding observation. Modified scans supervise point-level anomaly segmentation, and their original scans supervise normal-field prediction. Auxiliary anomalies and normal objects undergo the same road-placement and observation procedure. Normal cars, pedestrians, bicycles, and motorcycles retain their normal labels after insertion. The data therefore expose the model to both anomaly and normal decisions under the geometric changes introduced by object placement.

The paper develops two connected contributions. First, it formulates an input-excluded normal reference as shared three-dimensional return fields with analytic ray-conditioned distributions. Second, it introduces a hierarchical readout that keeps these normal hypotheses available to end-to-end point-level anomaly learning. The experimental study examines their contribution under fixed source supervision and evaluates transfer from nuScenes to STU using one score for all three official point-level metrics.

2 RELATED WORK

Point-level unknown detection. REAL learns unknown responses alongside known semantic classes, and LiON uses diverse synthetic objects to train point-wise abstention (Cen et al., 2022; Xu et al., 2025). Relative energy learning and neural distribution priors further connect anomaly supervision to learned segmentation scores (Li et al., 2026c;b). LIDO constructs complementary prototype, entropy, and contrastive evidence from normal training data (Mosco et al., 2026). COVAL studies visibility interventions for LiDAR anomaly learning (Yang et al., 2026). These methods establish semantic representation, auxiliary anomalies, and observation structure as useful sources of supervision. Our method develops an explicit conditional normal-return representation that a point-level decoder reads at inference.

Normal prediction for anomaly segmentation. Image resynthesis and inpainting provide normal references for road anomaly detection (Lis et al., 2024). DRAEM couples reconstruction with discriminative anomaly segmentation, while DenseHybrid combines normal-density and outlier-discrimination objectives (Zavrtanik et al., 2021; Grcić et al., 2022). InTra excludes the target image patch before predicting it from its neighborhood, and UniAD restricts reconstruction pathways to address input copying (Pirnay & Chai, 2022; You et al., 2022). In three dimensions, DOODLE augments LiDAR anomaly scores with diffusion-based feature reconstruction, and GPFlow models normal feature distributions with Gaussian prototypes (Oh et al., 2026; Li et al., 2026a). We predict distributions of measured distances from shared spatial hypotheses and retain their component structure in the anomaly readout.

Conditional prediction and geometric representation learning. Blind-spot denoising predicts a distribution from surrounding observations and combines it with the measured target (Laine et al., 2019; Krull et al., 2020). Conditional neural processes learn context-dependent prediction rules with uncertainty over queried outputs (Garnelo et al., 2018). In point clouds, masked reconstruction learns geometric representations (Pang et al., 2022); UniPAD and Ponder connect three-dimensional fields to rendered observations (Yang et al., 2024; Huang et al., 2023). Probabilistic LiDAR fields also connect ray attenuation with return distributions (McDermott & Rife, 2024). Our formulation brings

Model overview

Figure 1: Contextual normal return fields. The actual-observation encoder supplies point features. Block-excluded context predicts shared normal fields, which yield ray-conditioned distance distributions. A hierarchical readout combines their compatibility with the observed point into one anomaly score.

shared field prediction into the anomaly decision itself through a learned comparison with observed points.

Source-domain geometry and data construction. Scene completion, sensor-aware augmentation, and density variation support cross-domain LiDAR perception (Yi et al., 2021; Ryu et al., 2023; Kim et al., 2023). Mix3D and PolarMix vary the surroundings of real point-cloud instances while preserving their labels (Nekrasov et al., 2021; Xiao et al., 2022). We use source observations to construct both auxiliary anomalies and normal insertion controls, and organize the learning problem around their point-level semantic labels. This connects the training construction directly to the normal-reference mechanism.

3 CONTEXTUAL NORMAL RETURN FIELDS

3.1 POINT-LEVEL PREDICTION FROM OBSERVATIONS AND NORMAL CONTEXT

Let $X = \{(x_i, I_i)\}_{i=1}^n$ denote one LiDAR scan, where $x_i \in \mathbb{R}^3$ is a return coordinate and I_i its intensity. Each return has a direction d_i and distance r_i relative to the scan reference origin. The network produces a scalar $a_i \in \mathbb{R}$ whose ordering represents anomaly evidence. Training labels assign reliable normal points to zero, auxiliary anomaly points to one, and ambiguous points to an ignored set.

The model contains an actual-observation encoder and a normal-context predictor. The observation encoder uses LitePT-S (Yue et al., 2026) with a point-detail pathway to obtain $f_i \in \mathbb{R}^{64}$. It integrates multi-scale scene features, raw-point detail, and the point’s position within its voxel. The normal predictor partitions directions into angular blocks at three scales. For each target block B , it reads the surrounding blocks C_B and predicts a distributional reference for the target rays. Figure 1 reserves the model overview.

Points are encoded independently before their features are pooled within angular blocks. The target block is removed before any cross-block interaction. A context decoder then attends to the remaining neighborhood using learned hypothesis queries. Its outputs depend on the surrounding observations and the fixed block location and scale. At each scale, the same predicted field parameters serve all target rays in the block.

3.2 SHARED SPATIAL HYPOTHESES AND RETURN DISTRIBUTIONS

For a target block B , the context decoder produces K normal hypotheses with weights π_{Bk} . Each hypothesis is a positive spatial rate field formed by M anisotropic Gaussian components:

$$\pi_{Bk} = \text{softmax}(b_B)_k, \quad (1)$$

$$\lambda_{Bk}(x) = \sum_{m=1}^M a_{Bkm} \exp \left[-\frac{1}{2} (x - c_{Bkm})^\top \Sigma_{Bkm}^{-1} (x - c_{Bkm}) \right]. \quad (2)$$

The network predicts the centers c_{Bkm} , positive amplitudes a_{Bkm} , and full positive-definite covariances Σ_{Bkm} . The rate has units of inverse distance. Components within a hypothesis jointly

describe one spatial arrangement, and the mixture represents alternative arrangements supported by the context.

A ray queries this representation through $x_i(r) = o_i + rd_i$, with unit direction d_i . We model the distance of a recorded return within the interval $[L, U]$. Define the accumulated rate and its normalization by

$$H_{ik}(r) = \int_L^r \lambda_{Bk}(o_i + td_i) dt, \quad Z_{ik} = 1 - \exp[-H_{ik}(U)]. \quad (3)$$

The conditional return-distance density is

$$p_{ik}(r | C_B, d_i, \text{recorded return}) = \frac{\lambda_{Bk}(o_i + rd_i) \exp[-H_{ik}(r)]}{Z_{ik}}, \quad L \leq r \leq U. \quad (4)$$

The exponential term couples the rate at distance r to the accumulated rate at shorter distances. Normalization conditions the prediction on a recorded return in the interval. All target rays in B use the same λ_{Bk} , so a change to a spatial component affects several distance predictions through their ray geometry.

Restricting a three-dimensional Gaussian to a ray gives a one-dimensional Gaussian response. Consequently, H_{ik} is evaluated analytically through Gaussian cumulative probabilities. This provides differentiable distance densities without discretizing each ray into depth samples. Appendix A gives the derivation.

3.3 LEARNING A SHARED NORMAL EXPLANATION

Let $\mathcal{I}_B$ contain the reliably labeled normal returns within $[L, U]$ in block B of an original scan. The normal-learning objective assigns a common hypothesis to these returns:

$$P(R_B | C_B, Q_B, N) = \sum_{k=1}^K \pi_{Bk} \prod_{i \in \mathcal{I}_B} p_{ik}(r_i), \quad (5)$$

where Q_B contains the target ray directions and N denotes normal supervision. Thus, each component is trained to explain a collection of normal returns through one spatial field. The block loss is

$$\mathcal{L}_B = -\frac{1}{|\mathcal{I}_B|} \log \sum_{k=1}^K \pi_{Bk} \exp \left[\sum_{i \in \mathcal{I}_B} \log p_{ik}(r_i) \right]. \quad (6)$$

We average this loss over blocks containing normal targets, then over scales and reference scans. Point normalization gives each populated block a comparable contribution. This shared-hypothesis objective differs from independently mixing hypotheses for every normal point; that distinction forms one component comparison in Section 5.

3.4 HIERARCHICAL COMPATIBILITY WITH THE OBSERVED POINT

At inference, the decoder compares each actual return with the hypotheses predicted for its block. For scale s , let $(\mu_{ikm}^s, \tau_{ikm}^s, h_{ikm}^s)$ denote the center, width, and amplitude of component m after restriction to ray i . A shared map encodes these ray-specific parameters:

$$t_{ikm}^s = \phi[\mu_{ikm}^s/U, \log(\tau_{ikm}^s/U), \log(h_{ikm}^s U)] + \delta_s, \quad q_i = W_q f_i + \rho(r_i/U). \quad (7)$$

Here δ_s is a learned scale embedding, and the query q_i combines observed point features with measured distance. The first attention operation gathers component information separately within each hypothesis:

$$u_{ik}^s = \text{Attn}_m(q_i, \text{key}(t_{ikm}^s), \text{value}(t_{ikm}^s)). \quad (8)$$

The decoder augments this representation with the complete hypothesis density and its context weight, then aggregates the hypotheses:

$$v_{ik}^s = \psi[u_{ik}^s, \log p_{ik}^s(r_i), \log \pi_{Bk}^s], \quad (9)$$

$$e_i^s = \text{Attn}_k(q_i, \text{key}(v_{ik}^s), \text{value}(v_{ik}^s)). \quad (10)$$

This ordering preserves the grouping of Gaussian components into complete hypotheses. Both attention stages learn which predicted structures are relevant to the observed point. Field parameters remain the output of the block-excluded context predictor.

We additionally provide the point’s marginal normal log density,

$$\ell_i^s = \log \sum_k \pi_{Bk}^s p_{ik}^s(r_i), \quad a_i = g[f_i, e_i^1, e_i^2, e_i^3, \ell_i^1, \ell_i^2, \ell_i^3]. \quad (11)$$

The absolute density complements the attention-weighted features when all hypotheses assign little support to the return. The output a_i is the point-level anomaly score. The normal objective in Equation 5 shapes the shared representation during training; Equation 11 specifies its point-wise use during inference.

3.5 JOINT OPTIMIZATION

Detection supervision and normal prediction train the full model together. Let $\mathcal{L}_{\text{BCE}}$ be binary cross-entropy averaged within each present class and then across classes. We optimize

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha(t)(\tilde{\mathcal{L}}_{\text{AP}} + 0.1\tilde{\mathcal{L}}_{\text{AUC}} + 0.1\tilde{\mathcal{L}}_{\text{FPR95}}) + 0.1\mathcal{L}_{\text{normal}}. \quad (12)$$

The ranking terms operate on the same a_i across an effective batch of scans. They address the ordering of anomaly points, overall positive-negative separation, and the normal tail at high anomaly recall. Their weight $\alpha(t)$ increases from zero to one between 10% and 20% of the update budget. Appendix B defines the surrogates and their sampling. Detection gradients pass through the compatibility decoder and the return densities into the normal predictor, while original normal scans provide the likelihood supervision.

4 SOURCE TRAINING OBSERVATIONS

Backgrounds and semantic supervision. We construct training and development observations from the official nuScenes training and validation partitions (Caesar et al., 2020). They contain 700 and 150 scenes, respectively. A source point receives normal supervision when its annotated meaning agrees with the target normal set. Ambiguous source categories receive ignored labels. Camera observations support donor verification and local label refinement; model inputs are LiDAR coordinates and intensity.

Auxiliary anomalies and normal controls. The auxiliary object collection contains 40 verified instances, divided into 27 training and 13 development donors. They include bags, refuse containers, carts, toolboxes, and suitcases. STU defines its target anomalies separately from the nuScenes debris and pushable/pullable categories (Nekrasov et al., 2025); these source objects supply auxiliary anomaly supervision. Normal donors are adult pedestrians, bicycles, motorcycles, and cars. Both donor roles use the same placement procedure and retain their respective semantic labels.

Placement and recorded observations. Each placement fixes one original-size donor at a world pose supported by measured road points. Independent placements cover an unrestricted base placement and requested observation ranges of 20–30, 30–40, and 40–50 m. Vehicle motion then determines the sequence of viewing distances and directions. For each receiving scan, its recorded ray directions intersect the donor’s measured open surface. A closer supported intersection replaces the corresponding return. Observations are retained within the views supported by the measured donor surface. Background points in uncertain occlusion regions retain their coordinates with ignored supervision.

Paired learning records. Every modified observation is linked to its unmodified scan. The detection loss uses the modified points and their labels. The normal-field loss uses reliable normal returns from the original scan, including for records containing normal insertions. All versions of a scene remain in its original partition, and donor instance identities are partitioned with their source scenes. Table 1 summarizes the resulting records. Each original background scan appears once in the reading index.

Original scan, auxiliary anomaly insertion, normal insertion

Figure 2: Source supervision. Original observations provide normal-field targets. Independent auxiliary anomaly and normal-object insertions provide point-level detection labels through the same observation procedure. Panels are reserved for actual source examples.

Table 1: nuScenes source data. Modified records share their original scene and scan identities; the first two rows describe the underlying background observations.

	Training	Development
Background scenes	700	150
Original scans	28,130	6,019
Auxiliary anomaly donors	27	13
Anomaly insertion records	23,795	4,699
Normal insertion records	20,895	4,861
Total reading records	72,820	15,579

5 EXPERIMENTS

5.1 EVALUATION AND IMPLEMENTATION

Evaluation. The target benchmark is STU, with its public 19-sequence anomaly validation partition and 51-sequence hidden anomaly test partition (Nekrasov et al., 2025). The point-level evaluation uses valid labels in the 2.5–50 m interval and aggregates eligible scans containing at least five anomaly points. We report average precision (AP), area under the receiver operating characteristic curve (AUROC), and the false-positive rate at 95% anomaly recall (FPR95). All three metrics use the same checkpoint and unnormalized point score. Source development uses the nuScenes partition described in Section 4. Normal insertion observations additionally support an analysis of normal-point scores and false alarms.

Training. The observation backbone is initialized with official nuScenes LitePT-S weights. All model parameters are optimized jointly using AdamW, with peak learning rates of 2×10^{-4} for the backbone and 2×10^{-3} for the added modules. The fixed budget is 1,500 updates with eight scans per effective batch, corresponding to 12,000 scan visits. Full source-development evaluation follows the final update. The model uses 5 cm voxels, three angular scales of 1° , 2° , and 4° , four normal hypotheses, and eight Gaussian components per hypothesis. Computation uses FP32. Appendix C supplies the remaining optimization and architecture details.

5.2 BENCHMARK COMPARISON

Table 2 reserves the target comparison against LiDAR anomaly methods. The validation and hidden-test partitions are reported separately. The training resources associated with the published methods and the source-trained model are specified in Appendix D. The table records one operating model per method across AP, AUROC, and FPR95.

[Benchmark result analysis.]

5.3 CONTRIBUTION OF THE NORMAL REPRESENTATION

The component study uses a common source dataset, backbone initialization, optimization budget, and detection objective. A direct point classifier supplies the observation-only reference. A point-

Table 2: Point-level anomaly segmentation on STU. Values are percentages. Higher AP and AUROC and lower FPR95 are better. Empty cells reserve measurements for the initial draft.

Method	Validation			Hidden test		
	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$
NDP (Li et al., 2026b)						
REL (Li et al., 2026c)						
LIDO (Mosco et al., 2026)						
COVAL (Yang et al., 2026)						
Ours						

Table 3: Controlled component comparison on STU validation. Each row specifies a separately trained variant under the common source training setup.

Variant	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$
Direct point classifier			
Point-wise mixture objective			
Marginal-density readout			
Full model			

wise mixture variant replaces the block objective in Equation 5 with $\prod_i \sum_k \pi_k p_{ik}(r_i)$ while retaining the same field architecture and inference decoder. A marginal-density readout retains the observation feature f_i and the three log densities ℓ_i^s in the final head. The complete model additionally reads the individual hypotheses through Equations 8–10. These comparisons separate the contribution of a normal reference, shared-hypothesis training, and structured hypothesis readout.

[Component comparison results and interpretation.]

5.4 NORMAL CONTROLS AND SPARSE OBSERVATIONS

Normal insertion controls expose the model to cars and other normal objects under the same road-placement and observation changes as the auxiliary anomalies. Figure 3 reserves their score distributions alongside anomaly performance stratified by range and observed point count. An accompanying analysis groups the source results by donor instance and object type. These views relate the pooled point-level metrics to the diversity of observations represented in the data.

[Normal-control and sparse-observation findings.]

5.5 QUALITATIVE ANALYSIS

Figure 4 reserves examples of the observed point cloud, target annotation, direct-classifier output, and full-model output. For selected target rays, predicted distance distributions accompany the point scores. This presentation connects the learned normal reference to the final segmentation at object boundaries and under sparse observations.

[Qualitative findings.]

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Figure 3: Analysis of normal retention and sparse anomaly observations. Panels reserve normal insertion scores, range-stratified measurements, and measurements grouped by the number of observed object points.



Figure 4: Qualitative comparison on STU validation. Reserved panels show the input, annotations, point-classifier baseline, full-model scores, and normal-return distributions for the indicated rays.

6 CONCLUSION

We introduced contextual normal return fields for point-level LiDAR anomaly segmentation. The method predicts shared spatial hypotheses from block-excluded context, converts them into return-distance distributions through analytic ray integration, and reads their compatibility with observed points through a learned decoder. Original normal scans and modified source observations supply complementary supervision within one training objective. The formulation makes normal prediction available as a structured representation for the final anomaly decision.

[Empirical conclusion to be completed from the benchmark and component results.]

WRITING ASSISTANCE

AI tools assisted with literature organization, manuscript drafting, and formatting.

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A ANALYTIC GAUSSIAN INTEGRATION ALONG A RAY

Consider one component with covariance $\Sigma = AA^\top$ and rate

$$\lambda_m(x) = a \exp\left[-\frac{1}{2}(x - c)^\top \Sigma^{-1}(x - c)\right]. \quad (13)$$

For the ray $x(r) = o + rd$, define $v = A^{-1}d$ and $w = A^{-1}(o - c)$. Completing the square gives

$$\mu = -\frac{v^\top w}{v^\top v}, \quad \tau = (v^\top v)^{-1/2}, \quad \log h = \log a - \frac{1}{2}\|w + \mu v\|^2, \quad (14)$$

$$\lambda_m(o + rd) = h \exp\left[-\frac{(r - \mu)^2}{2\tau^2}\right]. \quad (15)$$

The integrated component rate is therefore

$$\int_L^b \lambda_m(o + rd) dr = h\tau\sqrt{2\pi} \left[\Phi\left(\frac{b - \mu}{\tau}\right) - \Phi\left(\frac{L - \mu}{\tau}\right) \right], \quad (16)$$

where Φ is the standard Gaussian cumulative distribution function. Summing these terms over m gives $H_{ik}(b)$. The conditional density integrates to one because

$$\int_L^U \lambda_{Bk}(o + rd) e^{-H_{ik}(r)} dr = \left[-e^{-H_{ik}(r)} \right]_L^U = 1 - e^{-H_{ik}(U)}. \quad (17)$$

All field and density operations are differentiable with respect to the predicted parameters. We compute interval probabilities in the log domain, using stable tail and narrow-interval expressions. Point queries are processed in chunks of 2,048.

The implemented observation model uses the scan reference origin and conditions on a recorded return in $[2.5, 50]$ m. Its learned rate is a representation for normal distance prediction. The conditioning event specifies which recorded measurements enter the likelihood objective. For returns outside $[L, U]$, log-density inputs to the readout are set to zero; these returns remain available as context.

B POINT-LEVEL TRAINING OBJECTIVES

Let $\mathcal{P}$ and $\mathcal{N}$ be anomaly and normal points across one effective batch. Binary cross-entropy averages the losses within each present class and then averages the class means. With both classes present,

$$\mathcal{L}_{\text{BCE}} = \frac{1}{2} \left[\frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \log(1 + e^{-a_p}) + \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \log(1 + e^{a_n}) \right]. \quad (18)$$

Ignored points remain available as input observations and are excluded from these supervised sets.

Ranking samples. We uniformly sample up to 256 anomaly anchors into $\mathcal{A} \subseteq \mathcal{P}$. All anomaly scores remain available when computing positive ranks and the soft recall threshold. Up to 4,096 normal points form $\mathcal{S}_-$. When $|\mathcal{N}| > 4,096$, this set contains the 512 highest-scoring normal points and 3,584 uniformly sampled points from the remainder. The former have weight one and the latter have weight $(|\mathcal{N}| - 512)/3584$. Otherwise all normal points are retained with unit weights. Denote these weights by w_n .

Average-precision surrogate. With temperature $\tau = 1$ and logistic function σ , the soft positive and negative ranks of anomaly anchor p are

$$R_p^+ = 1 + \sum_{q \in \mathcal{P} \setminus \{p\}} \sigma\left(\frac{a_q - a_p}{\tau}\right), \quad R_p^- = \sum_{n \in \mathcal{S}_-} w_n \sigma\left(\frac{a_n - a_p}{\tau}\right), \quad (19)$$

$$\tilde{\mathcal{L}}_{\text{AP}} = 1 - \frac{1}{|\mathcal{A}|} \sum_{p \in \mathcal{A}} \frac{R_p^+}{R_p^+ + R_p^-}. \quad (20)$$

Pairwise separation. The AUROC-oriented term uses a smooth positive-negative ordering loss:

$$\tilde{\mathcal{L}}_{\text{AUC}} = \frac{1}{|\mathcal{A}| |\mathcal{N}|} \sum_{p \in \mathcal{A}} \sum_{n \in \mathcal{S}_-} w_n \text{softplus} \left(\frac{a_n - a_p}{\tau} \right). \quad (21)$$

Normal tail at high recall. The soft threshold θ satisfies

$$\frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \sigma \left(\frac{a_p - \theta}{\tau} \right) = 0.95. \quad (22)$$

We differentiate this threshold implicitly and optimize the weighted normal exceedance:

$$\tilde{\mathcal{L}}_{\text{FPR95}} = \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{S}_-} w_n \sigma \left(\frac{a_n - \theta}{\tau} \right). \quad (23)$$

For $u_p = (a_p - \theta)/\tau$, the threshold derivative is $\partial\theta/\partial a_p = \sigma'(u_p)/\sum_q \sigma'(u_q)$. Ranking terms are zero when either class is absent. The BCE term continues to supervise the present class. The schedule is

$$\alpha(t) = \text{clip} \left(\frac{t/T - 0.1}{0.1}, 0, 1 \right), \quad T = 1500. \quad (24)$$

C ARCHITECTURE AND OPTIMIZATION DETAILS

The model contains 13,215,474 trainable parameters. Table 4 lists its dimensions. The 2° and 4° grids are nested coarsenings of the 1° grid. A target block reads the 24 other blocks in its 5×5 neighborhood at the selected scale. Empty neighbors use an attention mask; an entirely empty neighborhood uses a learned context token. The context predictor uses LayerNorm and block-local pooling.

Table 4: Implemented model dimensions. Shared maps use the same parameters across components, hypotheses, and angular scales.

Component	Configuration
Observation backbone	LitePT-S; 5 cm voxel size
Actual point state	64-dimensional scene, detail, and within-voxel terms summed
Context point encoder	$[x/U, y/U, z/U, I, d_x, d_y, d_z]$; $7 \rightarrow 64 \rightarrow 64$
Block representation	Mean/max pooling; 128 dimensions
Hypothesis decoder	Four queries; two four-head cross-attention layers; feed-forward width 256
Spatial fields	Four hypotheses, each with eight full-covariance Gaussian components
Ray-component map ϕ	$3 \rightarrow 16 \rightarrow 64$; 32-dimensional keys and values
Hypothesis map ψ	$34 \rightarrow 64 \rightarrow 64$; 32-dimensional keys and values
Compatibility attention	Four heads per stage; eight dimensions per head
Final point head	$163 \rightarrow 128 \rightarrow 64 \rightarrow 1$

The context and field decoder share parameters across scales. Covariances use Cholesky factors, and positive parameterizations give positive amplitudes. Hypothesis queries are initialized separately. Field centers are initialized across the valid distance interval in a coordinate basis determined by the target angular block. The actual-point query combines the 64-dimensional point state projected to 32 dimensions with a $1 \rightarrow 16 \rightarrow 32$ distance encoding. Hidden layers in the final head use normalization and GELU.

AdamW uses $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, and weight decay 0.005, with zero decay for biases and normalization parameters. Learning rates warm up for the first 5% of updates from 0.1 of their peak

value, then follow cosine decay to 0.01 of that value. The gradient norm is clipped at one. The fixed source run uses random seed zero and the preconstructed observations directly.

Source ordering reserves up to 20% of complete batches for observations whose anomaly-point median range is at least 30 m. Anomaly observations are grouped by powers-of-two point-count intervals. These batches contain one to four anomaly scans, at least one original normal scan, and at least one normal insertion scan. The procedure reorders records within a traversal. Its grouping variables are measured from the saved observations.

The effective batch contains eight scans. Scores from the batch are collected before computing the joint detection objective. Cached score gradients are then replayed through per-scan forward computations, with the normal-reference loss added to the same parameter update. Original reference scans require only the normal predictor. Normalization state advances once per training observation. This arrangement retains the effective-batch ranking objective while processing the backbone one scan at a time.

D TRAINING RESOURCES AND OBSERVATION COVERAGE

Table 5 records training resources alongside the target comparison. The model uses official nuScenes backbone initialization and nuScenes auxiliary objects. The designated source run uses nuScenes for optimization and endpoint development evaluation. STU validation is reported as a public development partition, and the hidden partition supplies the separate test evaluation.

Table 5: Resources associated with the benchmark comparison. Empty entries reserve information to be completed for the exact compared checkpoints.

Method	Backbone training sources	Auxiliary anomalies	Target-domain use
NDP			
REL			
LIDO			
COVAL			
Ours	nuScenes	nuScenes donors	Public validation reporting

The source background partitions contain disjoint acquisition logs, and donor instance identities follow the source partition. The 40 auxiliary donors originate from 21 training and 12 development scenes. Their single-observation point counts range from 8 to 601. Normal donors come from adult pedestrians, bicycles, motorcycles, and cars. Original debris and pushable/pullable labels remain ignored; verified donor surfaces receive auxiliary anomaly labels after insertion. Reliable normal supervision includes compatible source labels and 4,752 manually refined building points across six source frames.

Every placement uses the donor at its measured size and a fixed world pose. A support point lies at least one meter inside the convex hull of local measured road points, and collision checks use scene observations. The receiving scan supplies the queried directions. Surface intersections replace recorded slots when they are nearer and supported by donor visibility. All independent placement versions remain with the same original scene. Table 6 gives counts derived from the resulting observations.

Object-type analysis accompanies pooled point-level evaluation. Refuse containers contribute 44.7% of training anomaly points and 86.8% of development anomaly points. Source development therefore includes donor-wise and type-wise summaries, together with range and point-count strata. The original scans, insertion records, and donor identities define the units for these summaries.

E ADDITIONAL RESULTS

[Additional source-development and computational results.]

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Table 6: Observed source coverage. Range statistics use the saved returns. The five-point condition follows the point-level evaluation eligibility rule.

	Training	Development
Successful auxiliary placements	1,828	366
Successful normal placements	1,542	359
Scans with at least five anomaly points	10,486	1,796
Scans with one to four anomaly points	10,180	1,985
Auxiliary anomaly points in range	235,996	52,341
At least five anomaly points beyond 30 m	494	223
At least five inserted normal points beyond 30 m	575	129

Table 7: Source-development evaluation and computational measurements. Runtime entries will specify the evaluation hardware and include both model pathways.

Model	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$	Time per scan
Direct point classifier				
Full model				